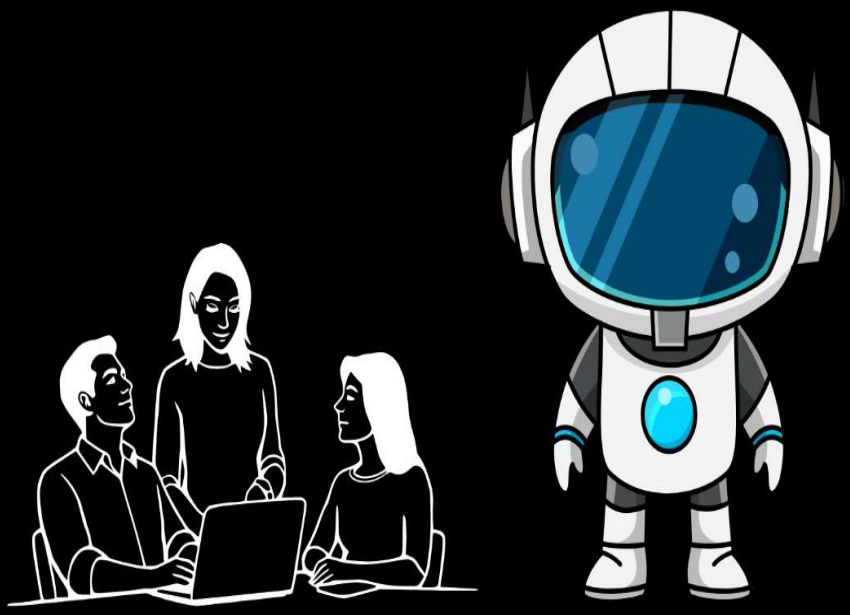


AI IMPACT

ON JOB SECTOR

EDA Questions:

Q1:What is the distribution of Productivity_Change_%?¶

Q2: What is the distribution of Job Satisfaction?

Q3: How are employees distributed across AI_Adoption_Level (Low, Medium, High)?

Q4:Is there any missing value in data?

Q5: What is Connectivity/Relationship between AI-Adoption and Productivity Change?

Q6: Which industries have the highest number of employees?

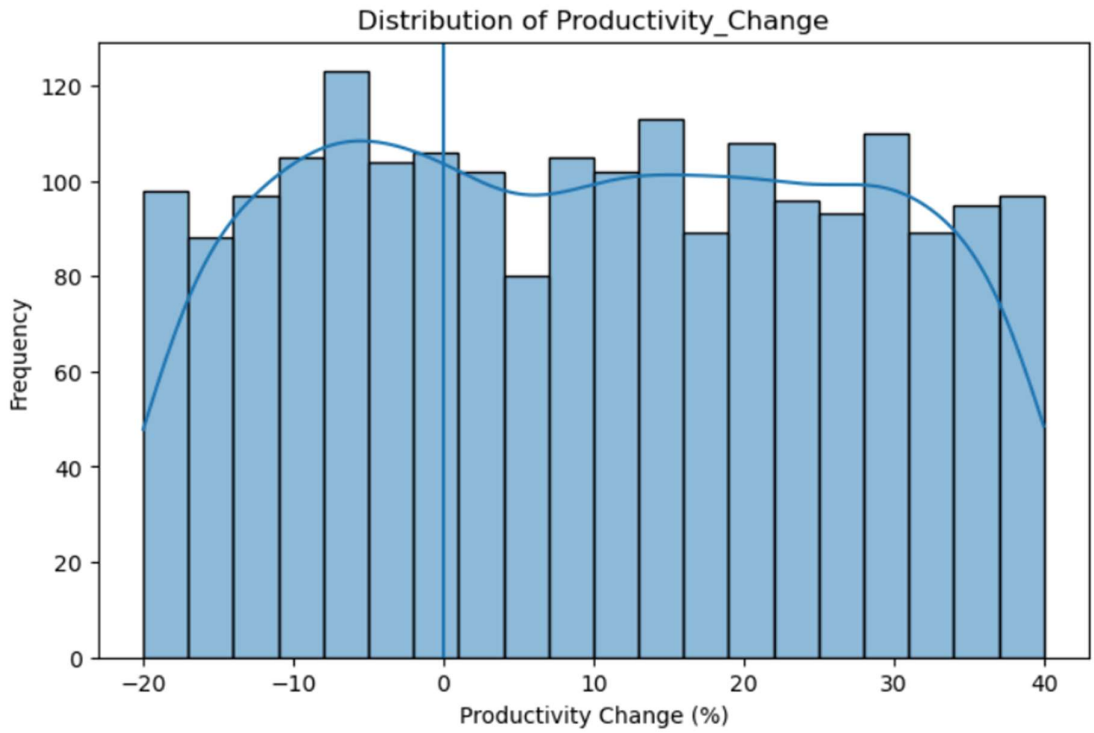
Q7:How many employees fall into each Automation_Risk category?

Q8: How many Employees work remotely?

Q9: What is the average value of Job_Satisfaction by AI_Adoption_Level

Q10: What is the distribution of Work_Hours_Per_Week?

Q1:What is the distribution of Productivity_Change_%?



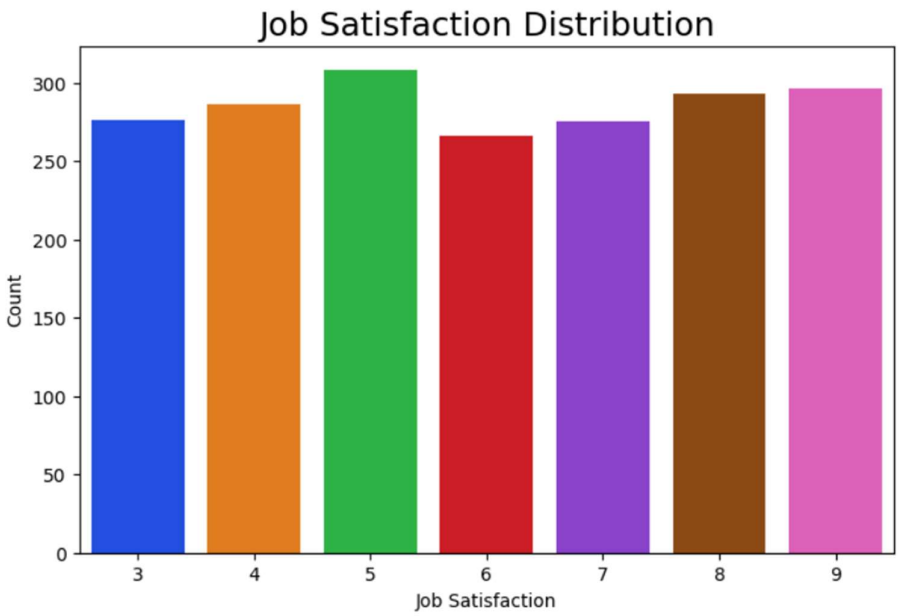
As we can see it is roughly symmetrical.

Conclusion:

The histogram shows that productivity change is spread on both positive and negative sides,with most values near ZERO.This mean AI has a mixed effect__it improve productivity for some Employess but also decrease some Employess productivity.The overall change of AI on Productivity is very small, which means AI doesn't have a big impact on Productivity Change

"Most employees show small changes in productivity, while fewer employees experience large increases or decreases."

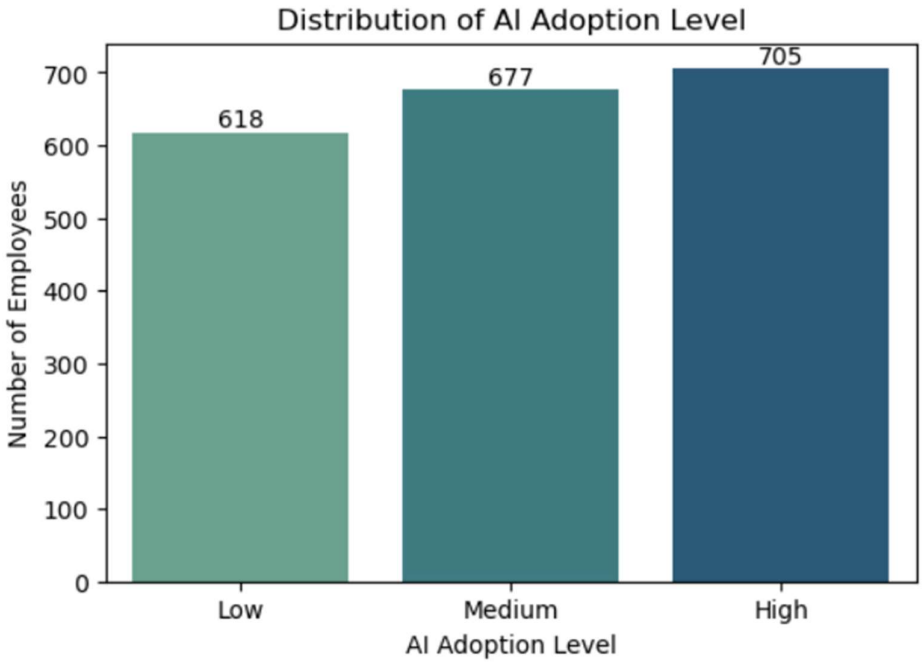
Q2: What is the distribution of Job Satisfaction?



Conclusion:

Employees show a mix of satisfaction levels, but overall most are moderately to highly satisfied with their jobs.

Q3: How are employees distributed across AI_Adoption_Level (Low, Medium, High)?



Conclusion:

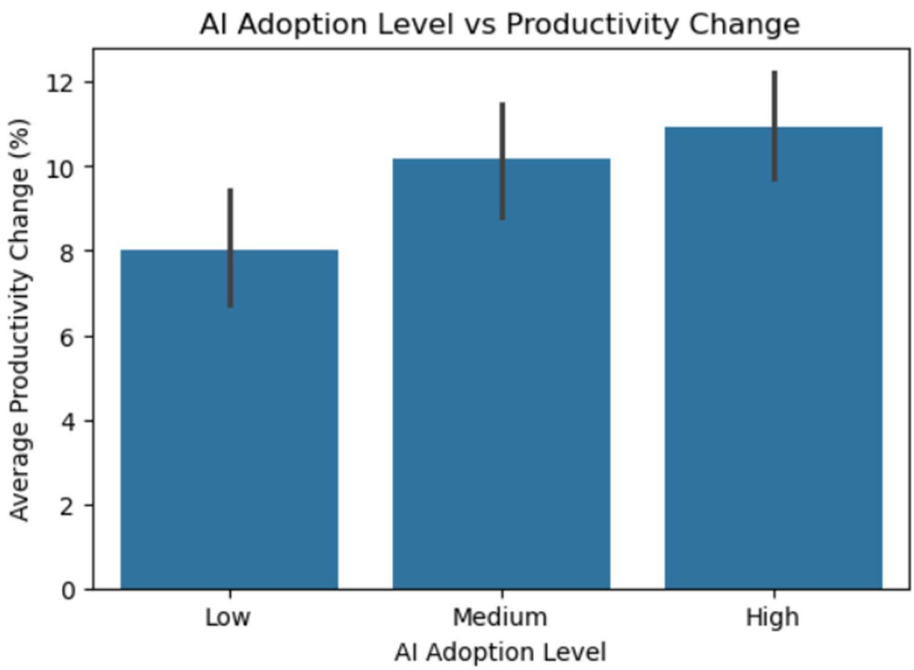
AI adoption is quite balanced across employees, with slightly more people using AI at a high level. This means AI is commonly used, but not all employees are using it at the same level.

Q4:Is there any missing value in data?

```
df.isnull().sum().sum()
```

```
np.int64(0)
```

Q5: What is Connectivity/Relationship between AI-Adoption and Productivity Change?

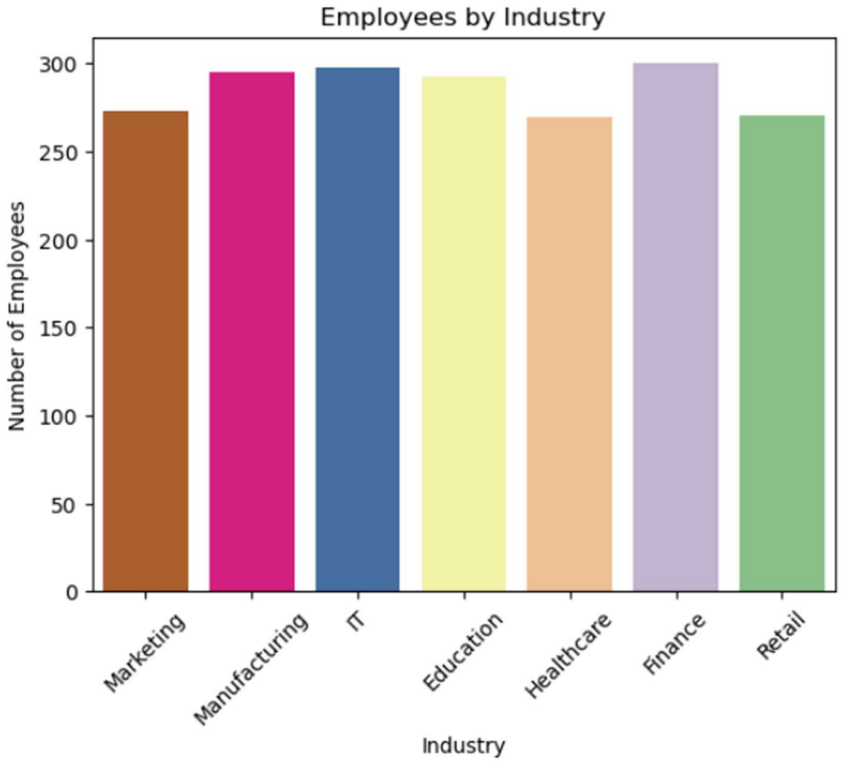


Conclusion:

- Employees who use more AI tend to have higher productivity

- There is a positive relationship between AI adoption and productivity

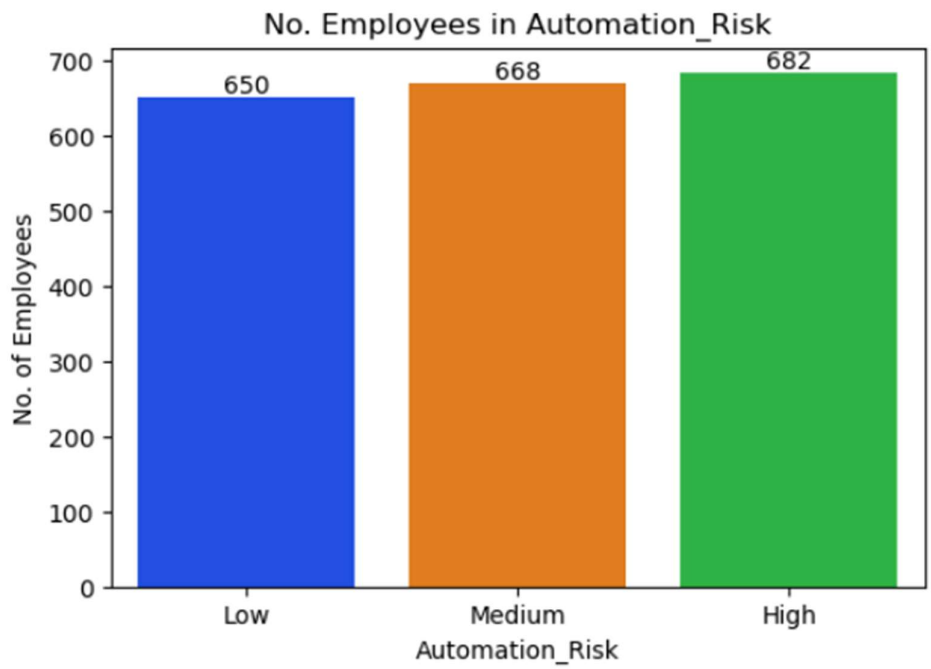
Q6: Which industries have the highest number of employees?



Conclusion:

As we can see all industries have good number of Employees but Finance industry slightly has the highest num of Employees.

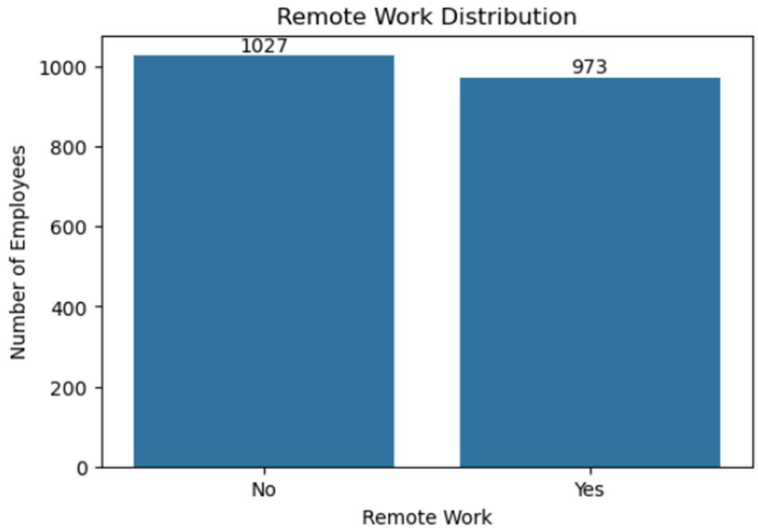
Q7:How many employees fall into each Automation_Risk category?



Conclusion:

- Many jobs are affected by automation, suggesting the need for employee upskilling
- Without upskilling, employees in higher-risk roles may face job uncertainty due to automation.

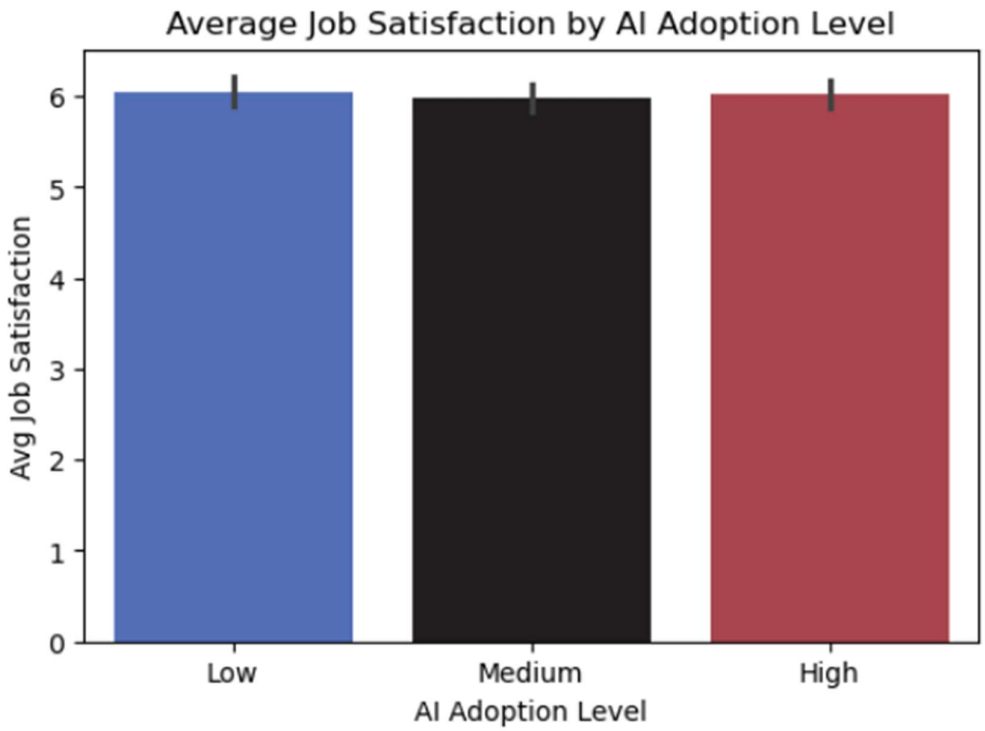
Q8: How many Employees work remotely?



Conclusion:

Employees are almost evenly split between remote and office work, indicating a hybrid work environment.

Q9: What is the average value of Job_Satisfaction by AI_Adoption_Level?



Conclusion:

- Job satisfaction remains almost the same across all levels of AI adoption.
- Increasing AI usage:

❌ Does NOT reduce satisfaction

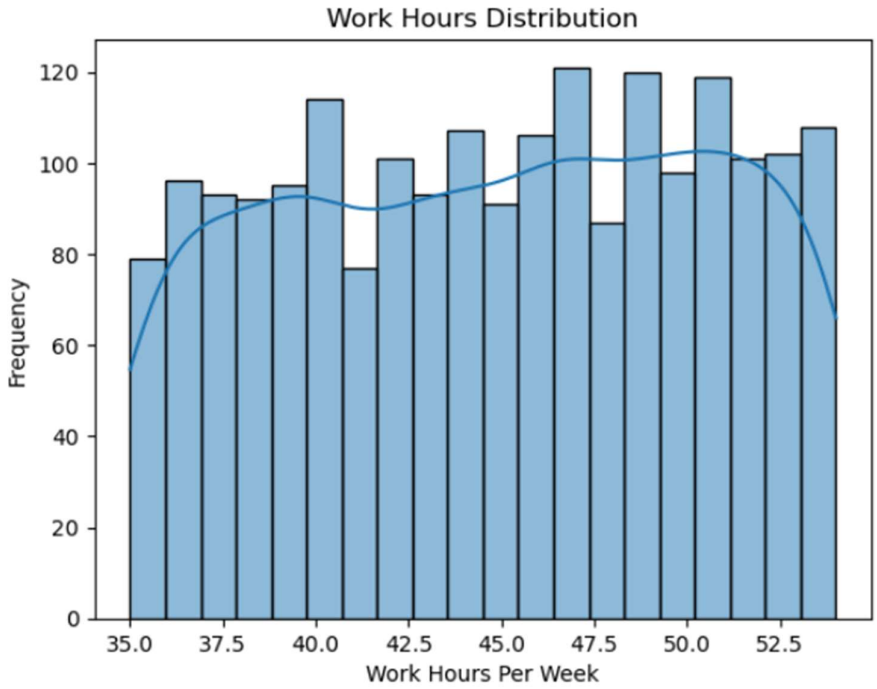
❌ Does NOT significantly increase satisfaction

So AI has neutral impact on employee satisfaction

Most Imp:

- "AI improves productivity without negatively affecting employee satisfaction."

Q10: What is the distribution of Work_Hours_Per_Week?



Conclusion:

Work hours are almost balanced.

Key Insights:

- Productivity remains stable for the majority, indicating that external factors (like AI) do not drastically impact everyone equally.
- Overall employee sentiment is **generally positive**, but there is still a segment that may require attention.
- AI usage is **widely spread across the workforce**, indicating growing adoption rather than limited usage.
- AI appears to have a **positive impact on productivity**, especially when actively used.
- Using AI tools may reduce workload or improve efficiency, leading to better employee satisfaction.
- A significant portion of employees may be **affected by automation in the future**, highlighting the need for skill development.
- Workload is **balanced**, and there is no strong evidence of widespread overworking.
- The company operates in a **hybrid work model**, offering flexibility in work arrangements.
- Balanced work hours and flexible work arrangements may contribute to **stable productivity levels**.

Final Summary:

The organization is transitioning toward AI adoption while maintaining stable productivity and satisfaction levels. However, the presence of moderate to high automation risk highlights the importance of upskilling employees to remain competitive in an evolving work environment.

Business Recommendations:

I. Upskill employees
II. Improve tool usage

III. Monitor productivity
IV. Support low-satisfaction employees
V. Strengthen hybrid work model